



Bangladesh University of Business and Technology

Course Title: Computer Networks Lab

Course Code: CSE 320

ASSIGNMENT

Assignment Name: Compare CSMA/CA and CSMA/CD

Assignment No.: 01

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Introduction

Carrier Sense Multiple Access (**CSMA**) is a network access protocol used at **the data layer** to reduce data collisions on shared channels. This protocol to improve efficiency, commonly used in Ethernet and Wi-Fi.

There are two variants of **CSMA**, each designed to handle how devices manage potential data collisions on a shared communication channel. These types differ based on how they respond to the detection of a busy network:

- **CSMA/CA (Collision Avoidance):** Used in wireless networks (e.g. **Wi-Fi**) to avoid collisions by having stations wait for a random, extra time even after the channel becomes idle.
- **CSMA/CD (Collision Detection):** Mainly used in wired networks (e.g. **Ethernet**), it detects collisions and stops transmission immediately to save bandwidth.

CSMA/CD and **CSMA/CA** are essential protocols for managing network traffic and minimizing collisions in shared communication channels.

Types of CSMA Access Modes

There are 4 types of access modes available in **CSMA**. It is also referred to as 4 different types of CSMA protocols which decide the time to start sending data across shared media.

1-Persistent: It senses the shared channel first and delivers the data right away if the channel is idle. If not, it must wait and continuously track for the channel to become idle and then broadcast the frame without condition as soon as it does. It is an aggressive transmission algorithm.

Non-Persistent: It first assesses the channel before transmitting data; if the channel is idle, the node transmits data right away. If not, the station must wait for an arbitrary amount of time (not continuously) and when it discovers the channel is empty, it sends the frames.

P-Persistent: It consists of the 1-Persistent and Non-Persistent modes combined. Each node observes the channel in the 1-Persistent mode and if the channel is idle, it sends a frame with

a P probability. If the data is not transferred, the frame restarts with the following time slot after waiting for a ($q = 1 - p$, $q = 1 - p$ probability) random period.

O-Persistent: A supervisory node gives each node a transmission order. Nodes wait for their time slot according to their allocated transmission sequence when the transmission medium is idle.

CSMA/CA

CSMA/CA is a MAC layer protocol that helps wireless devices avoid collisions by ensuring that only one device transmits at a time. **CSMA/CA** is used in wireless networks (e.g., **Wi-Fi**), where detecting collisions after they happen is difficult due to the broadcast nature of wireless signals. Instead, **CSMA/CA** tries to prevent collisions from occurring in the first place:

- **Carrier Sense:** The device checks if the channel is free before transmission.
- **Multiple Access:** As with **CSMA/CD**, multiple devices may want to transmit data simultaneously.
- **Collision Avoidance:** To prevent potential collisions, **CSMA/CA** uses techniques like a backoff timer and Request to Send (**RTS**) / Clear to Send (**CTS**):
 - **Backoff Timer:** If the channel is busy, the device waits for a random period before rechecking.
 - **RTS/CTS:** A device requests permission to transmit by sending an **RTS** signal. If the receiving device is ready, it replies with a **CTS**. This exchange helps prevent other devices from starting transmissions during this time.

Usage of **CSMA/CA**:

- **CSMA/CA** is the foundational protocol for **Wi-Fi** networks (**IEEE 802.11**), ensuring that devices do not interfere with each other.
- Wireless networks need **CSMA/CA** due to the “*hidden node*” problem, where devices can’t detect each other’s signals and could unintentionally collide.

Wireless channels are highly susceptible to interference and overlapping signals. **CSMA/CA** offers a way to manage the shared medium more effectively, allowing devices to avoid collisions by confirming that the channel is clear before and during transmission.

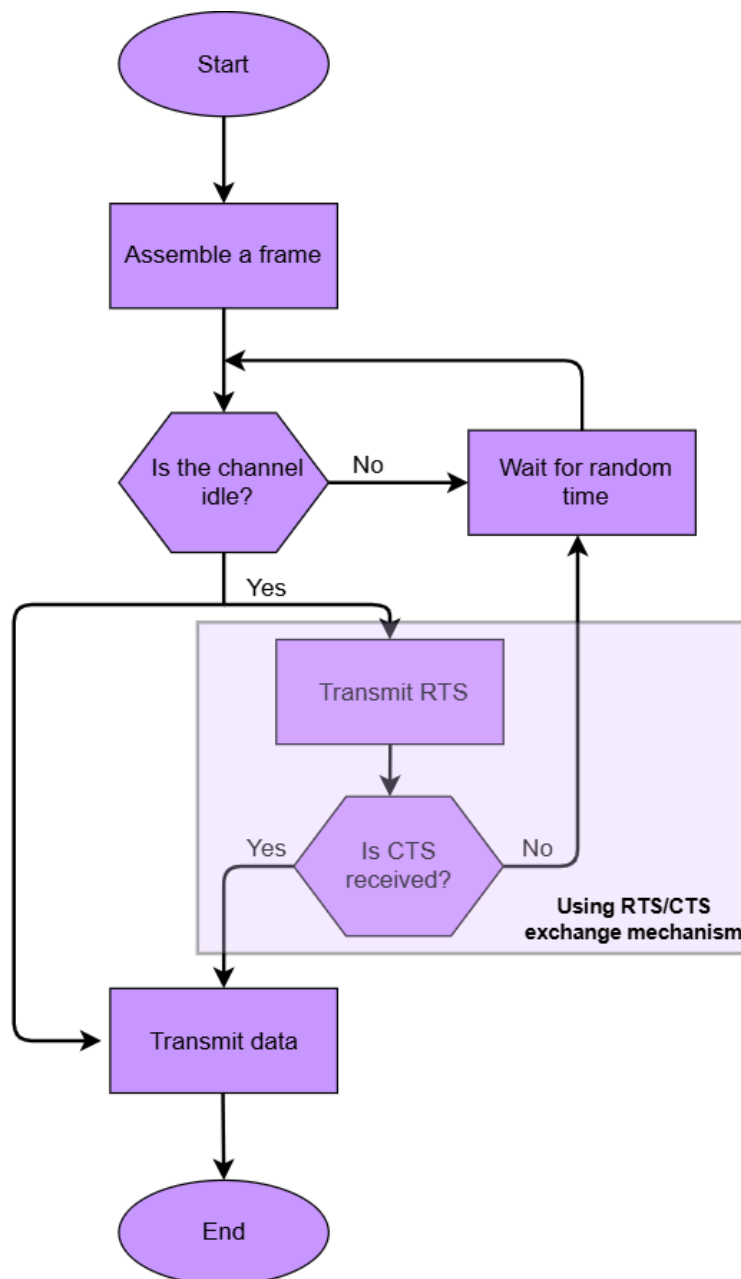


Figure 1.1: CSMA/CA workflow with RTS/CTS exchange mechanism.

Working principle of CSMA/CA:

- First, the stations sense the transmission medium. This implies that the carrier sense keeps a close eye on the radio channel and determines whether other stations are transmitting at the same time.
- If the transmission medium is already in use, a random backoff is initiated, during which the station waits until a new check begins for a determined period. The same happens at all other stations that are not actively broadcasting or receiving. It only

occurs if the station is unaware that the medium is already occupied (which is set by the wireless listening stations to tell a station how long it must wait before accessing the wireless medium).

- If the network is available, the station initiates, which regulates how long a station waits before commencing a media transmission. Before t , which is the amount of time a station should wait before sending its request frame, the channel is examined thoroughly. If it remains free, a random backoff begins, and the optional exchange starts, which minimizes frame conflicts caused by the hidden node problem. If the request to send has been received successfully at the receiver side and there hasn't been a collision, the frame grants the sender permission to use the transmission medium.
- All other stations are alerted that the network is in use. As a result, they raise their network allocation vector again and delay making another attempt to verify if the channel is free.
- The station then begins the transmission. The receiver confirms that the data packet has been successfully received after waiting for the t , which is the time required to process a data package, and then sends an ACK (acknowledgement) frame to indicate to the sender that the data frame has been received. It also sets the network allocation vector to 0, indicating that the network is now open for a new transmission.

Advantages of CSMA/CA

The following points discuss the advantages of using CSMA/CD:

- For collision detection, CSMA/CD is better than CSMA.
- It is used for quick collision detection on a shared channel.
- Compared to CSMA/CA, it has lesser overhead.
- Using CSMA/CD helps prevent the waste of transmission in any way.
- Whenever mandatory, it consumes or shares all of the bandwidth.

Disadvantages of CSMA/CA

Disadvantages are given as follows:

- After reaching the maximum distance, collision detection is impossible.
- Not ideal for larger networks as collision detection is difficult to detect.
- The performance gets considerably disrupted when more devices are added.

CSMA/CD

CSMA/CD is primarily used in Ethernet networks, particularly traditional wired Ethernet setups. In CSMA/CD, devices on the network listen to the channel and only begin transmitting if the channel is free. Here is how it works:

- **Carrier Sense:** Before sending data, a device checks if another device is already transmitting on the channel.
- **Multiple Access:** Since multiple devices share the channel, there's potential for simultaneous transmissions.
- **Collision Detection:** If a collision occurs, both devices stop transmitting immediately. The collision is detected using a method such as monitoring voltage levels on the line.
- **Backoff:** After detecting a collision, each device waits for a random period (called backoff) before attempting to retransmit. This helps avoid repeated collisions by staggering retransmission times.

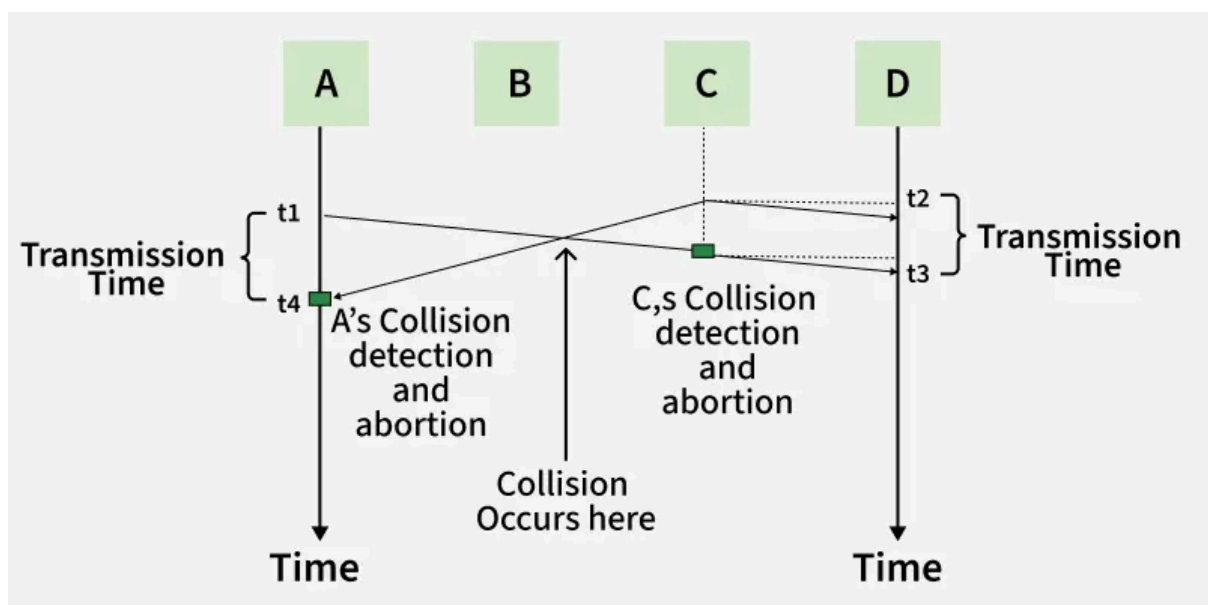


Figure 1.2.: Collision in CSMA/CD.

Algorithm of CSMA/CD:

- The station that intends to transmit the data detects if the carrier is available or busy by perceiving its state. If a carrier is empty, the transmission starts.
- The condition $T(t) \geq 2Tp$ where $T(t)$ is the transmission delay and $T(p)$ is the propagation delay, is used by the transmission station to identify collisions.
- As soon as it notices a collision, the station sends the jamming signal.
- Following a collision, the transmitting station interrupts transmission and waits for a predetermined period, known as the "*back-off time*". The station then re-transmits the signal.

Although CSMA/CD is more efficient than CSMA, a few considerations are discussed below relating to its effectiveness:

- The efficiency of CSMA/CD decreases with increase in distance.
- Even if efficiency improves with a larger packet size, there is still a limit as to how it can be. The packets can only be a maximum of **1500 bytes** long.

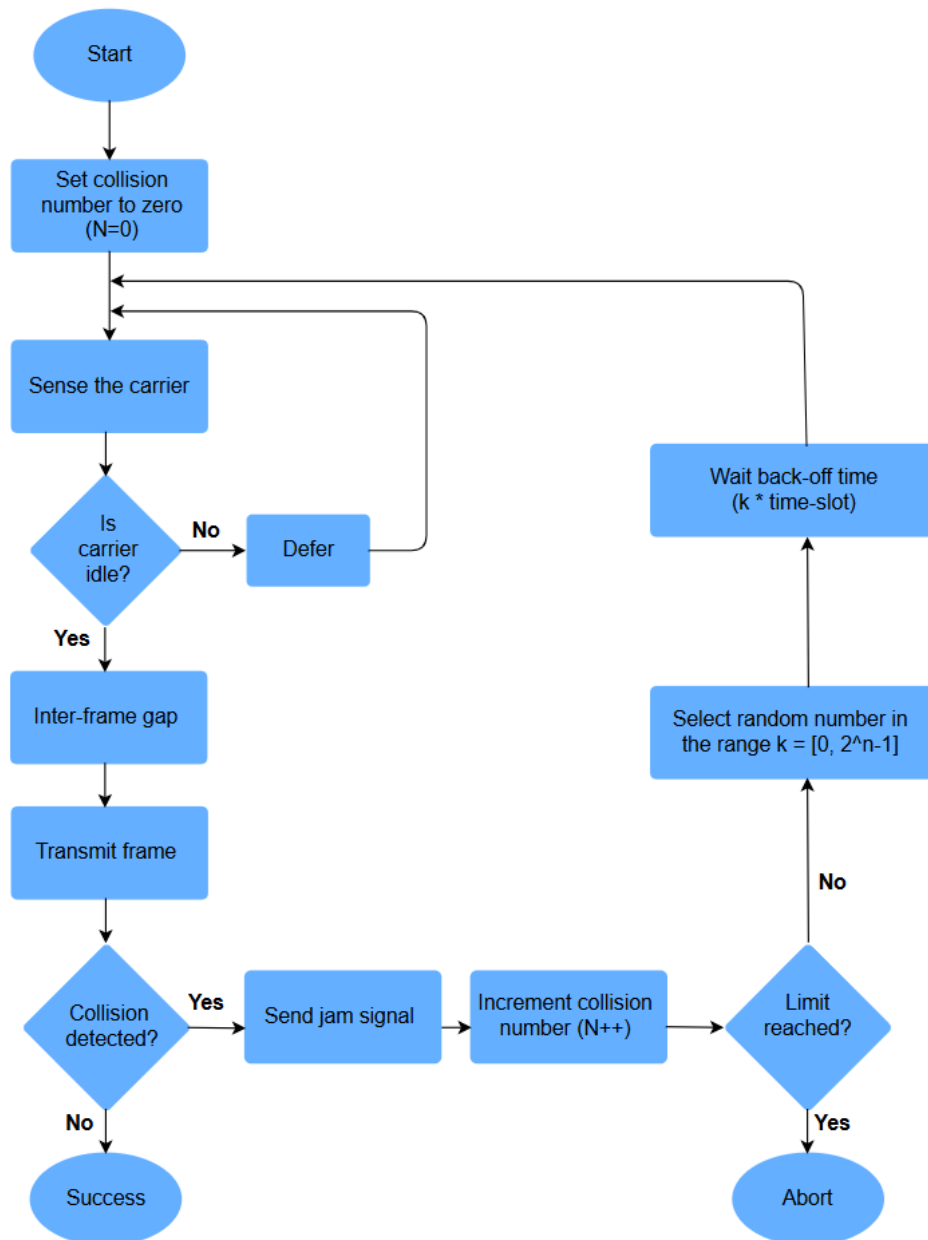


Figure 1.3.: CSMA/CD procedure.

Usage of CSMA/CD:

- Wired Ethernet networks, especially older **10BASE5** and **10BASE2** Ethernet standards, rely heavily on **CSMA/CD**.
- In full-duplex Ethernet or switched Ethernet environments, **CSMA/CD** is less relevant, as collisions are largely avoided by dedicated communication paths.

CSMA/CD became a standard in wired networks because it effectively manages collisions, enhancing efficiency by allowing devices to share a medium without constantly interfering with each other.

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Accessing Common Channels

Carrier Sense Multiple Access (**CSMA**) is a network protocol used to manage shared communication channels by ensuring devices “listen” to the medium before transmitting data, a concept known as “sense before transmit” or “listen before talk”. By detecting if the channel is already in use, nodes aim to avoid collisions, although they can still occur due to propagation delays.

Comparison of CSMA/CD and CSMA/CA Protocols

Protocol	Transmission Behavior	Collision Detection Method	Efficiency	Use Cases
CSMA/CD	Monitors medium after sending a frame, retransmits if necessary.	Collision detection by monitoring transmissions.	High	Wired networks with moderate to high traffic.
CSMA/CA	Monitors medium while transmitting, adjusts behavior to avoid collisions.	Collision avoidance through random backoff time intervals.	Low	Wireless networks with moderate to high traffic and high error rates.